THE SUPPORT OF A QUASI-COHERENT MODULE PULLS BACK ALONG AN ETALE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a single stalk comparison; the faithful flatness of strict henselization is what upgrades the tensor equality to the equivalence of vanishing.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Stalks of quasi-coherent sheaves on an algebraic space through an etale presentation. Properties of Spaces, Lemma spaces-properties-lemma-stalk-quasi-coherent, used at proof lines 3031-3035. Relates the stalk of the pullback to the stalk of F.
- Etale local rings as strict henselizations of local rings of schemes. Properties of Spaces, Lemma spaces-properties-lemma-describe-etale-local-ring, used at proof lines 3036-3039. Identifies the ring map appearing in the stalk comparison.
- Strict henselization is faithfully flat. More on Algebra, Lemma more-algebra-lemma-dumb-properties-henselization, used at proof lines 3040-3042. Turns the tensor equality into an equivalence of vanishing statements.
- The definition of the support as the set of points with nonzero stalk. used at proof lines 3043-3044. Concludes the equality of the two supports.

Source reference. Original lemma 07QM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9837–9837. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04KF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5971–5971. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7558–7558. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let U be a scheme and let $\varphi : U \to X$ be an étale morphism. Then*

$$Supp(\varphi^*\mathcal{F}) = |\varphi|^{-1}(Supp(\mathcal{F}))$$

where the left hand side is the support of $\varphi^*\mathcal{F}$ as a quasi-coherent module on the scheme U.

Proof. Let $u \in U$ be a (usual) point and let $\overline{x}$ be a geometric point lying over u. By Properties of Spaces, Lemma 05VP we have $(\varphi^*\mathcal{F})_u \otimes_{\mathcal{O}_{U,u}} \mathcal{O}_{X,\overline{x}} = \mathcal{F}_{\overline{x}}$. Since $\mathcal{O}_{U,u} \to \mathcal{O}_{X,\overline{x}}$ is the strict henselization by Properties of Spaces, Lemma 04KF we see that it is faithfully flat (see More on Algebra, Lemma 07QM). Thus we see that $(\varphi^*\mathcal{F})_u = 0$ if and only if $\mathcal{F}_{\overline{x}} = 0$. This proves the lemma. □